CPC353 Assignment 1:

Special Topic Study on Natural Language Processing

2026/2027

Released Date: 21 October 2026 (Wednesday)
Submission Date: 13 November 2026 (Friday)
Submission method: Softcopy in Word and Pdf (eLearn)

This project is a 3-in-a group project.

1. Assignment Overview

Domain: Code-switching in English–Malay Food Product/Eatery reviews

Objective**:** To investigate how English and Malay are mixed within online food product/eatery reviews using text analytics techniques. You will construct a text dataset, identify and analyse code-switching patterns, and interpret the linguistic characteristics of the collected reviews.

Code-switching refers to the use of two or more languages within the same conversation, sentence, or text. In Malaysian online reviews, for example, a reviewer may write:

- “Sedap gila, definitely worth the price!”
- “The chips are crunchy tapi agak mahal.”
- “Packaging okay, but rasa dia kurang fresh.”

These examples illustrate how reviewers may alternate between English and Malay words or phrases to express opinions, describe product attributes, and convey emphasis.

You are required to formulate a research question, construct a dataset, apply at least **two NLP techniques**, and discuss the findings in relation to code-switching behaviour.

Note: For those not familiar with Malay language, you may switch to other languages.

2. Formulate Questions

Select one main research question from the following:

1. What are the common English–Malay code-switching patterns found in food product reviews?
2. Which English and Malay words or phrases frequently occur in code-switched reviews?
3. How are code-switching patterns associated with sentiment, such as positive or negative product evaluations?
4. Which product attributes, such as taste, price, texture, and packaging, are frequently discussed in code-switched reviews?
5. How do informal expressions, abbreviations, and spelling variations affect the identification and analysis of English–Malay code-switching?

The selected research questions should determine the choice of NLP techniques and the scope of the analysis.

3. Dataset Construction

You are required to collect a dataset of online food product (eatery) reviews containing English, Malay, or a mixture of both languages.

3.1 Data Collection

- Collect approximately 100–200 review texts. You may use LLM to generate review texts from a seed set of real data. AI generated texts should be indicated as Source: AI (LLM Used) All real data should be indicated as Source: URL (URL of the product/eatery).
- Focus on one product or product category, such as snacks, instant noodles, biscuits, beverages, or other packaged food products. Likewise for eatery.
- Collect reviews from publicly accessible sources, such as online shopping platforms, food review websites, or public discussion forums.
- Include the source URL and relevant metadata, such as product category, rating, and review date, where available.
- Ensure that the dataset contains sufficient examples of English–Malay code-switching for meaningful analysis.
    - At minimum, classify each review into one of the following categories:
    - English: Primarily English, without identifiable Malay–English switching.
    - Malay: Primarily Malay, without identifiable English–Malay switching.
    - Code-switched: Contains both English and Malay within the same review.
    - Other/uncertain: Contains another language, insufficient context, or ambiguous language use.

3.2 Data Preparation

- Remove duplicate reviews and identify missing or empty entries.
- Preserve the original review text for linguistic analysis.
- Define the unit of analysis, such as a complete review, sentence, or language-switching instance.
- Record the number of reviews collected and the number retained after cleaning.

4. Text Analysis

Select and apply at least one technique relevant to code-switch analysis.

- Concordances
- Stop Words
- Part of Speech
- N-gram
- Grammar (Phrase/Structure)
- Semantics

Example

Concordance Analysis

- *Generate concordance lines for frequently occurring English or Malay words.*
- *Examine the words and phrases surrounding selected terms.*
- *Identify how language switching is used in different contexts.*
- *Develop a small bilingual lexicon of recurring expressions, informal spellings, or mixed-language phrases.*

5. Report Your Findings

You will need to submit the report in softcopy format. Minimum 3 pages (max 6 pages) (excluding cover). The report needs to follow manuscript (IEEE format)

Template: http://www.ieee.org/conferences_events/conferences/publishing/templates.html

Guideline:

The report needs to include minimally these sections. (These are guidelines only; you may modify the outlines accordingly to improve your report)

- Title and Abstract
- Overview (Motivation and Question)
- Scenario (this part contains the dataset description)
- Related Methods (this part contains the related works on the selected methods)
- Text Exploration (this part contains your own analysis using the selected methods on your sample data)
- Findings (report the findings)
- Conclusions
- References

EVALUATION CRITERIA (Refer to rubric document for the criteria)

1. Understanding of selected topic & scenario.
2. Selection of the literatures (timeliness, quality)
3. Conciseness of summary for related works.
4. Dataset preparation and reusability.
5. Text analysis details, suitability and findings.
6. References.
7. Overall writing, structure, flow, and language.